

This week classes: Tu (Th) (F) 2pm No class tomorrow (Wed)
First HW out today, due next Tu before class

Review Solving $Ax=b \leadsto$ Row reduce Augmented matrix A/b

- Consistent $\Leftrightarrow$ No pivot in last col of (echelon form) of A/b
- When consistent, "dim" of solution space = # free variables
= # columns - # pivots
= $C - \# \text{ pivots}$

Observation. # pivots in any $A \leq \min(r, c)$
will use: either equality is interesting

- $Ax=0$ (homog system) has only the trivial soln ($x=0$) $\left. \begin{array}{l} \Rightarrow \text{Every column of (echelon form of) } A \text{ has a pivot} \end{array} \right\} \begin{array}{l} \Rightarrow \\ C \leq r \end{array}$
- $Ax=b$ has a solⁿ for every possible b $\left. \begin{array}{l} \Rightarrow \text{Every row of "A" has a pivot} \end{array} \right\} \Rightarrow r \leq c$

e.g. ① 3 eqⁿ in 5 unknowns $\leadsto A$ is 3×5
 $\leadsto Ax=0$ will always have
"2-dim" solⁿ set

($\Rightarrow Ax=b$ for any b will be either inconsistent -OR-
its solⁿ set will have the same "dimension" as
the —// of $Ax=0$ # free vars)

② 5 eqⁿ in 3 vars $\leadsto A$ is 5×3 , again # pivots ≤ 3

$\Rightarrow$ Last 2 rows of any echelon form of A must be all 0's

$\Rightarrow$ There will be vectors b for which $Ax=b$ is inconsistent

(3) 5 eqⁿ & 5 vars $\# \text{ pivots} \leq 5$

$\# \text{ pivots} = 5 \Leftrightarrow$ each col has a pivot $\Leftrightarrow Ax=0$ has unique solⁿ

$\Leftrightarrow$ each row has a pivot $\Leftrightarrow Ax=b$ consistent for any b

$\Leftrightarrow \text{RREF}(A) = \text{Identity}(5 \times 5) \text{ matrix}$

Observe: for square matrix A ,

$Ax=0$ has unique solⁿ (namely 0) $\Leftrightarrow Ax=b$ has a solⁿ for each b (which must be unique)

Problem: Show that any two of the following implies the third for an $r \times c$ matrix A

- A is square i.e. $r=c$
- $Ax=0$ has only the trivial solⁿ
- $Ax=b$ is consistent for all b .

Function associated to a given matrix A ($r \times c$)

$$\mathbb{R}^c \xrightarrow{f} \mathbb{R}^r$$

$$x \mapsto Ax \quad \text{i.e. } f(x) = Ax$$

- $Ax=b$ has a solution $\Leftrightarrow b \in \text{range of } f$

$Ax=b$ has a solⁿ for all $b \Leftrightarrow f$ is onto

- f is one to one $\Leftrightarrow f(x) = Ax = b$ has at most one solⁿ for any b

proof of $\Leftarrow$ $\Leftrightarrow Ax=0$ has only the trivial solⁿ

proof of $\Leftarrow$

If $Ax_1 = Ax_2 = b$, then $A(x_1 - x_2) = b - b = 0$, so $x_1 - x_2$ is a

solution to $Ax=0 \Rightarrow x_1=x_2$

Observe $\{b \mid Ax=b\}$ has a solution = range of function $x \mapsto Ax$

This is a ~~subset~~ subspace of $\mathbb{R}^r$

$\{x \mid Ax=0\} = \text{"kernel of } f\text{"} = \text{"null space of } f\text{"}$
Def $= \text{"null space of } A\text{"}$

This is a ~~subset~~ subspace of $\mathbb{R}^c$

Observe $f\left(\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}\right) = \begin{bmatrix} a_{11} & \dots & a_{1c} \\ a_{21} & & \\ \vdots & & \\ a_{r1} & & a_{rc} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{r1} \end{bmatrix} = \text{1st column of } A$

$e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ i^{th} std

$Ae_i = i^{\text{th}} \text{ column of } A$

Recall defⁿ of a v. space V

- have addition, subtraction, scalar multiplication
- some properties hold

Subspace of (real) v. space V is a subset W s.t.

- W is itself a v. space under the same operations

$\Leftrightarrow$ • W is closed under $+$, $\cdot$ (i.e. $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$
(Desired properties hold automatically in W because they hold in V) $\{cw_1 \in W$

example ① Subspaces of $\mathbb{R}^2$ are $\{0\}$, $\mathbb{R}^2$, any line thro' the origin. **Ex** There are all the subspaces of

② $\mathbb{R}^2$. Do the same for $\mathbb{R}^3$

Exercise For any $r \times c$ matrix A and the associated function f , $\text{range}(f)$ is a subspace of $\mathbb{R}^r$, $\text{ker}(f)$ is a subspace of $\mathbb{R}^c$.